

SYNOPSIS

Title

FoodBridge – Real-Time Surplus Food Rescue & Routing Platform

Introduction

Food waste is one of the most persistent inefficiencies in the food service industry: restaurants, cafés, and grocery stores routinely prepare or stock more than they sell, and the surplus that remains at closing time is discarded rather than redirected to people who need it. At the same time, local shelters, food banks, and community kitchens operate on tight budgets and limited donated supply, often unaware that usable surplus exists just a few streets away. FoodBridge is a web-based platform designed to close this gap by connecting food donors with nearby recipient organizations in real time. Rather than relying on manual phone calls and informal arrangements, the system automates the discovery, matching, and logistics of surplus food redistribution — turning what would otherwise be waste into a delivered meal before it expires.

Problem Statement

Restaurants, cafés, and grocery stores routinely discard unsold food at closing time — not because it is unwanted, but because coordinating same-day pickup with a shelter or food bank is logistically difficult: phone calls go unanswered, and by the time a match is found, the food has already expired. Recipient organizations, meanwhile, have no visibility into what surplus exists nearby at any given moment, so viable donations are missed entirely. FoodBridge closes this gap with a platform that lets donors post surplus in real time, automatically matches each post to the nearest recipient with sufficient capacity, and routes deliveries by urgency — prioritizing whatever will expire soonest.

Objectives

- Enable donors to post surplus food listings in real time with minimal effort.
- Automatically match surplus listings to the nearest recipient organization with sufficient capacity.
- Prioritize matching and delivery based on expiration urgency to minimize spoilage.
- Coordinate volunteer-based pickup and delivery logistics through the same platform.
- Provide administrators with tools to verify participating organizations and resolve disputes.
- Track and report measurable impact — meals rescued and food waste diverted — for donors and the platform as a whole.

Scope

FoodBridge is scoped as a city-level coordination platform connecting three user groups — donor businesses, recipient organizations, and volunteer drivers — within a defined metropolitan service area. The system covers the full donation lifecycle: listing surplus, automated matching, volunteer dispatch, delivery confirmation, and impact reporting. It does not cover food safety certification or regulatory compliance verification, which remain the responsibility of participating organizations, nor does it process financial transactions, since donations are provided free of charge. The platform is built as a solo-developer academic project and deployed as a functional web application accessible through a standard browser; a native mobile application is treated as a future extension rather than part of the initial build.

Proposed Solution / Methodology

FoodBridge follows a client-server architecture with a Django REST Framework backend, a PostgreSQL/PostGIS database, and a React frontend. The workflow begins when a donor creates a surplus listing specifying item type, quantity, an expiration window, and a pickup window. The system geocodes the donor's address using GeoPy and stores it as a spatial point in PostGIS. A matching engine then queries nearby recipient organizations within a configurable radius, filters by remaining capacity and dietary or storage constraints, and ranks candidates using a

weighted score that combines geographic proximity with time-to-expiry, so that listings closest to spoiling are surfaced and matched first. Once a recipient accepts a match, the system assigns an available volunteer and generates a pickup-and-delivery route. Donors, recipients, and volunteers are notified at each stage — new match, pickup confirmed, delivery completed — via SMS and email. All completed transactions are logged and aggregated into an analytics layer that powers the impact-reporting dashboard. Role-based authentication (Donor / Recipient / Volunteer / Admin) governs access to each part of the system, and an administrative interface allows platform staff to verify new organizations and resolve disputes.

Features/Modules

- Donor Module – surplus listing creation and management
- Recipient Module – organization profiles, capacity limits, and dietary/storage constraints
- Matching Engine – automated proximity, capacity, and urgency-based matching
- Prioritization Queue – expiration-based ranking of active listings
- Volunteer Dispatch Module – route assignment for multi-stop pickup and delivery
- Notification System – real-time SMS/email alerts across all roles
- Admin Panel – organization verification and dispute resolution
- Analytics Dashboard – impact metrics such as meals rescued and food waste diverted

Technology Stack

- Frontend: React, Tailwind CSS
- Backend: Django, Django REST Framework (Python)
- Database: PostgreSQL with PostGIS extension
- Geocoding: GeoPy (OpenStreetMap / Nominatim)
- Notifications: Twilio (SMS), Django's built-in email framework (SMTP)
- Authentication: Django's built-in auth system, djangorestframework-simplejwt
- Hosting/Deployment: Vercel (frontend), Render (backend and database)

System Requirements

Hardware: Standard laptop or desktop with at least 8 GB RAM (16 GB recommended for running the local database and development servers together).

Software: Python 3.11+, Node.js 18+, PostgreSQL 15+ with the PostGIS extension, and Git.

Browser: Latest two versions of Chrome, Firefox, Edge, or Safari, with JavaScript enabled.

Operating System: Cross-platform — Windows, macOS, or Linux for development; end users need only a modern web browser on any device.

Expected Outcomes

- A fully functional web platform demonstrating real-time, automated matching between food donors and recipient organizations.
- A measurable reduction in food waste for participating donor businesses.
- Improved and more consistent access to surplus food for shelters and food banks.
- A significant reduction in the manual effort currently required to coordinate food donations.
- A working impact dashboard donor businesses can reference for their own sustainability reporting.
- A solo-built, deployed full-stack application demonstrating multi-role access control, geospatial algorithms, and third-party service integration.

Future Scope

- Native mobile applications for donors and volunteers, with push notifications and live GPS tracking.

- AI-based surplus forecasting to help recipient organizations anticipate incoming donations.
- Direct integration with point-of-sale/inventory systems so surplus is detected and listed automatically.
- Expansion to multiple cities or regions with dynamic service-area configuration.
- Gamification and recognition features to encourage consistent donor and volunteer participation.

Conclusion

FoodBridge addresses a real and recurring inefficiency — usable food discarded simply because donors and recipients lack a fast way to find each other — with a solution grounded in automation rather than manual coordination. By combining geospatial matching, urgency-based prioritization, and integrated logistics into a single platform, the project delivers genuine value to donor businesses, recipient organizations, and the communities they serve, while providing a technically substantial build — spanning role-based access control, an automated matching algorithm, and real-time notifications — suitable for a solo capstone project.